Geometric Cosmology with a Regular Bounce and Transient Stiff-Era Expansion: An Einstein–Cartan Sound-Horizon Reduction for H_0, a Background S_8 Shift, and Open Growth Physics

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The standard ΛCDM model faces persistent tensions in the inferred Hubble constant H_0 and in the amplitude of late-time structure growth (S_8), while its classical extrapolation toward the earliest epochs leads to an initial singularity. We investigate a minimal Einstein–Cartan–Kibble–Sciama extension in which the intrinsic spin density of matter induces an effective spin-fluid correction to the Friedmann dynamics: the resulting repulsive spin term regularizes the high-density regime through a non-singular bounce. A transient stiff-like phase before recombination is then implemented through a *separate, calibrated positive branch*; the microphysical bridge between the two remains open (PO-F1-SIGMA0). In a calibrated implementation, the effective amplitude $F_{\rm ion} = 1.2765$ sets the sound-horizon reduction, giving $r_s \simeq 135.65$ Mpc and $H_0 \simeq 73.04$ km s^{-1} Mpc^{-1} at fixed acoustic angle θ_* (held fixed at $\Omega_m = 0.315$; under the internal closure $\Omega_m = \omega_m/h^2$ the same background lowers θ_* by $\simeq 6.5\%$, so $H_0 = 73.04$ is conditional; open problem PO-F1-THETA, whose resolution requires an Einstein–Cartan perturbation solver not expected in the near term). At fixed physical densities the higher H_0 lowers $\Omega_m = \omega_m/h^2$, which strongly reduces S_8: a CAMB ΛCDM reference on this background gives $S_8 \simeq 0.780$, against $\simeq 0.831$ on the Planck background; the weak-lensing value $S_8 = 0.766$ is not an ECF prediction, the residual correction depending on the ECF perturbative dynamics, which is not implemented. In the historical configuration, the calibrated solution improves the effective statistic by $\Delta\chi^2 \simeq -39.5$ (ΔAIC $\simeq -35.5$, ΔBIC $\simeq -23.8$, very strong on the Kass–Raftery scale; historical, $\Omega_m = 0.315$; not recomputed: no ECF perturbation solver yet). The same ansatz gives $t_0 \simeq 13.33$ Gyr for $w = -1$ and $\Omega_m = \omega_m/h^2 = 0.267$, compatible with the globular-cluster age bound; the residual background tension lies in the BAO distances and in the DESI (w_0, w_a) comparison, and remains open. We identify observational tests — cosmic birefringence, high-ℓ CMB damping-tail residuals, a contracted acoustic ruler, and a blue-tilted primordial gravitational-wave background — and make the scripts regenerating every published figure and number publicly available.

Version note. *Preprint v8 (October 2026; internal revision 7.1), aligned with the Extended version (internal revision 13; concept doi:10.5281/zenodo.19577447), to be deposited alongside it. This compact PRD-style summary accompanies the full extended manuscript and is prepared for journal submission. Changes since v1: two illustrative (non-data) figures removed and the text condensed for the journal format; birefringence stated as the falsifiable $\beta \neq 0$ prediction with 0.35° as a calibrated consistency check; H_0 labelling clarified (ECF calibration 73.04; external H0DN anchor $H_0 = 73.50 \pm 0.81$ km s^{-1} Mpc^{-1}, A&A 708, A166, 2026, consistent at 0.6σ). Numerical results ($\Delta\chi^2 = -39.5$ [historical, $\Omega_m = 0.315$, not recomputed: no ECF perturbation solver yet], $H_0 = 73.04$, $r_s = 135.65$ Mpc).*

I. INTRODUCTION

The ΛCDM cosmological model has achieved remarkable success in describing the cosmic microwave background, baryon acoustic oscillations, large-scale structure, and the late-time expansion history [5]. Nevertheless, the present observational landscape contains statistically persistent tensions. The Hubble tension compares early-universe inferences of H_0, typically near 67–68 km s^{-1} Mpc^{-1} within ΛCDM [5], with local distance-ladder determinations near 73–74 km s^{-1} Mpc^{-1} [6]. A second tension concerns the amplitude of structure growth: weak-lensing surveys tend to prefer lower values of S_8 than those inferred from the CMB under the same baseline model [7, 8]. Finally, the classical Friedmann solution of general relativity is past singular, indicating the breakdown of the effective description in the high-density primordial regime.

This paper asks whether these three issues may have a common geometric origin. Einstein–Cartan gravity relaxes the torsion-free assumption of general relativity by

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allowing spacetime torsion to be sourced by the intrinsic spin of matter [9–11]. Although such effects are negligible under ordinary late-time conditions, they become relevant at extreme spin densities. The resulting repulsive spin contribution has long been known to provide a possible route toward singularity avoidance through a cosmological bounce [12–14].

We study a specific calibrated realization of this idea, referred to as the Einstein–Cartan Framework (ECF). The aim of this short manuscript is to isolate the physical mechanism, summarize the joint phenomenological fit, and state the observational tests that would support or disfavor the construction. A complete derivation of the modified Friedmann dynamics, numerical implementation, dataset choices, and extended statistical tables is given in the companion *Foundation I: The Metric Universe — Extended Version* [2], and the broader program in the PIT document [1]; the present article remains self-contained at the level of its mechanism, benchmark calibration, and falsification tests.

II. SPIN-TORSION COSMOLOGY

Einstein–Cartan theory extends general relativity by introducing torsion $T^{\lambda}_{\mu\nu}$, algebraically sourced by spin density. For a homogeneous and isotropic universe described by an effective Weyssenhoff spin fluid [15], the background dynamics can be written in the schematic Friedmann form [2, 13, 14]

$$H^2 = \frac{8\pi G}{3}\left(\rho_r + \rho_m + \rho_\Lambda - \rho_{\rm spin}\right), \tag{1}$$

where $\rho_{\rm spin} = 2\pi G\sigma^2$ denotes the effective spin-density contribution and σ^2 is the averaged spin-density magnitude. The negative sign of $\rho_{\rm spin}$ is not a convention but a direct consequence of the ECKS field equations: the spin-spin contact interaction between fermions is *repulsive*, entering the effective stress-energy tensor of the Weyssenhoff fluid with opposite sign to the energy densities of radiation, matter, and Λ.

The physical consequence is immediate and fundamental. Since $H^2 \geq 0$ by definition, the right-hand side can never become negative. As the scale factor a decreases toward the past, $\rho_{\rm spin}$ grows as a^{-6} (see Eq. (2) below) while radiation grows only as a^{-4}. There therefore exists a unique minimum scale factor $a_{\rm min}$ at which

$$\rho_r + \rho_m + \rho_\Lambda \;=\; \rho_{\rm spin}(a_{\rm min}) \quad\Longrightarrow\quad H = 0.$$

This is the **bounce condition**: $H = 0$ at finite density, with $\dot{H} > 0$ immediately after. The universe cannot contract further — the repulsive spin pressure halts the collapse and reverses the expansion. This is not a fine-tuned cancellation: because $\rho_{\rm spin} \propto a^{-6}$ always overtakes $\rho_r \propto a^{-4}$ as $a \to 0$, the bounce is a generic consequence of the ECKS coupling, independent of initial conditions. The argument for robustness against spin inhomogeneities rests on the locality of the interaction (Poplawski's analyses [13, 14] are FLRW, i.e. assume homogeneity): the spin-spin contact interaction is local, so spatial fluctuations average out on super-Hubble scales before the bounce epoch. Classical GR-based bouncing scenarios generally require null-energy-condition violation in spatially flat FLRW [30]; the ECKS mechanism does not evade this requirement but carries it in a local, algebraic spin–torsion term rather than in an additional exotic field [13, 14]. The shear problem is not thereby solved: shear scales as a^{-6}, the same power as the effective torsion term, so the two compete on equal footing. Popławski finds the anisotropic bounce to hold only when particle production dominates over shear [31], isotropisation arising from the post-bounce era rather than from torsion itself. We make no claim of resolving the BKL problem. At late times, the a^{-6} suppression ensures complete decoupling, the residual spin density falling well below any Solar System observational threshold.

Conservation of the spin density gives the characteristic scaling

$$\rho_{\rm spin} \propto a^{-6}. \tag{2}$$

This component therefore grows more rapidly toward the past than radiation or matter. With the sign convention of Eq. (1), it acts as a repulsive contribution in the high-density limit.

The first consequence is singularity regularization. As the scale factor decreases, the spin term can balance the positive radiation and matter components, yielding $H \to 0$ at a finite minimum scale factor $a_{\rm min}$. The classical Big Bang singularity is then replaced by a non-singular bounce within the effective theory.

The second consequence is a transient stiff-like post-bounce era. Because the spin contribution scales as a^{-6}, it is rapidly diluted, but before it becomes negligible it modifies the early expansion history. In the ECF implementation, this transient phase is the mechanism that can reduce the sound horizon while leaving late-time cosmology close to the standard background.

a. Scope of Eq. (1). That equation describes the ultradense ECKS regime, where the unpolarised spin-spin contact term is repulsive and enters subtractively — the form that produces the bounce above. The calibrated background used for the sound-horizon reduction below is a *distinct*, effective description: the residual axial condensate is treated as a positive stiff component, $E^2(z) = \Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_{\rm spin,0}(1+z)^6 + \Omega_\Lambda$, which is what raises $H(z)$ before recombination and hence lowers r_s. Its positive sign, stiff equation of state and isotropic reduction are calibrated properties of that branch, not consequences derived from Eq. (1); the relation between the two regimes is an open problem (PO-F1-SIGMA0). All numerical results quoted here derive from the effective branch. See Foundation I Extended [2], §*Sign convention*, for the full discussion.

III. JOINT PHENOMENOLOGICAL FIT

A. Sound-Horizon Reduction

The CMB inference of H_0 is tied to the sound horizon at the baryon drag epoch,

$$r_s(z_{\rm drag}) = \int_{z_{\rm drag}}^{\infty} \frac{c_s(z)}{H(z)}\, dz. \tag{3}$$

An early-time increase in $H(z)$ reduces the time available for acoustic propagation and can lower r_s. If the angular acoustic scale is preserved through the corresponding distance calibration, a smaller sound horizon permits a larger inferred H_0; this holds at fixed Ω_m (here 0.315), and not under the internal closure $\Omega_m = \omega_m/h^2$ (PO-F1-THETA, below).

In the calibrated solution summarized here, the effective spin sector reaches a spin-to-radiation ratio of $\Omega_{\rm spin}^{\rm peak} \simeq 0.093$ at the calibration redshift $z_{\rm ref} = 5600$, and is rapidly diluted by recombination (0.36% at $z \simeq 1100$). This needs the a^{-6} law up to $z \sim 10^5$; extended to $T \sim 1\,\mathrm{MeV}$ the branch is excluded by BBN (PO-ECF-BBN-SPIN). This produces a reduction of the comoving sound horizon by about 7.7%, giving

$$r_s^{\rm ECF} \simeq 135.65\ \mathrm{Mpc}, \tag{4}$$

and a corresponding value

$$H_0 \simeq 73.04\ \mathrm{km\ s^{-1}\ Mpc^{-1}}. \tag{5}$$

These numerical values are inherited from the extended analysis [2] and should be interpreted as the calibrated point of the effective model rather than as parameter-free predictions. The value of H_0 assumes θ_* (held fixed at $\Omega_m = 0.315$; under the internal closure $\Omega_m = \omega_m/h^2$ the same background lowers θ_* by $\simeq 6.5\%$, so $H_0 = 73.04$ is conditional; open problem PO-F1-THETA, whose resolution requires an Einstein–Cartan perturbation solver not expected in the near term). The H_0 Distance Network (H_0DN) Collaboration has since confirmed $H_0 = 73.50 \pm 0.81\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [29], a community consensus result at 7.1σ tension with flat ΛCDM and at 5.0σ with BBN+BAO (DESI DR2) [28, 29]; the ECF calibration at $73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is consistent with this value at 0.6σ.

In the notation of the extended manuscript, the benchmark is organized by the common effective amplitude

$$F_{\rm ion} = 1.2765. \tag{6}$$

This parameter fixes the sound-horizon contraction. In earlier versions it also set the amplitude of the growth suppression described below, through a second calibrated coupling $\gamma_{\rm spin}$; that joint role is not established under the internal closure $\Omega_m = \omega_m/h^2$ (Sec. III B).

B. Structure-Growth Suppression

The same transient spin-torsion sector may also modify the effective growth history. In the phenomenological implementation of earlier versions, the growth coupling $\gamma_{\rm spin} = 0.3116$ was calibrated so that the clustering amplitude reproduces

$$S_8 \simeq 0.766, \tag{7}$$

the value preferred by weak lensing [7, 8]; that calibration used the Planck-background reference $S_8 = 0.832$ and is retained as historical only. With $\Omega_m = \omega_m/h^2 \simeq 0.267$, a CAMB ΛCDM reference on the same background gives $\sigma_8 \simeq 0.825$ and $S_8 \simeq 0.780$ (against 0.831 on the Planck background); this is a conditional reference, not an ECF prediction, since the early ECF physics that sets $r_s = 135.65$ Mpc is not included. The value $\sigma_8 \simeq 0.8122$ quoted in earlier versions is the target S_8 converted at $\Omega_m = 0.267$, not a computed amplitude. The residual $0.766/0.780 - 1 \simeq -1.8\%$ depends on the ECF perturbative dynamics, which is not implemented; within the phenomenological form it would correspond to $\gamma_{\rm spin} \simeq 0.066^{+0.069}_{-0.094}$, compatible with zero and not a determination. The intended physical interpretation — an effective backreaction of the spin-driven early phase on small-scale growth, preserving the large-scale acoustic structure — is therefore not tested quantitatively by the present chain. A complete perturbation-level treatment remains an important next step.

For compactness, the short paper represents this effect through the same phenomenological regulator used in the extended analysis,

$$G_{\rm eff}(k) = G_N\, T^2_{\rm tors}(k), \qquad T^2_{\rm tors}(k) = 1 - \frac{A_{\rm sup} k^2}{k^2 + k_{\rm cut}^2}, \tag{8}$$

with the former phenomenological values

$$A_{\rm sup} = 0.15, \qquad k_{\rm cut} = 0.10\, h\,\mathrm{Mpc^{-1}}. \tag{9}$$

At leading order this gives $G_{\rm eff} \simeq G_N[1 - A_{\rm sup}(k/k_{\rm cut})^2]$ for $k \ll k_{\rm cut}$. Equation (8) should be understood as a controlled phenomenological compression of the spin-torsion backreaction rather than as a substitute for the full perturbative derivation. Under the internal closure, $A_{\rm sup} = 0.15$ is not a calibration on $S_8 = 0.766$: propagated into σ_8 on the CAMB reference above it would give $S_8 \simeq 0.740$, and it is not to be combined with $\gamma_{\rm spin}$, which would count the same suppression twice; the growth-suppression mechanism remains open.

C. Statistical Summary

The extended companion analysis evaluates the calibrated solution (no global Markov-chain fit was run) using a custom Python numerical suite and a combined set of early- and late-universe constraints, including Planck 2018, SH0ES, DES weak lensing, and BAO data [2]. The present paper reports the resulting benchmark solution.

For this benchmark combination, the calibrated ECF solution improves the effective global statistic by

$$\Delta\chi^2 \simeq -39.5 \tag{10}$$

TABLE I. **Effective χ^2 comparison for the benchmark dataset combination.** The values summarize the statistical accounting used in the companion extended analysis. They should be read as an internal comparison for the adopted likelihood construction rather than as a replacement for a full Boltzmann-code likelihood analysis. Historical values at $\Omega_m = 0.315$, not recomputed. [c]The BAO row belongs to that historical accounting; with $\Omega_m = \omega_m/h^2 \simeq 0.267$ the four-bin diagonal BAO comparison (BOSS DR12 + eBOSS DR16) gives $\chi^2 = 12.11$ (ECF) against 1.79 (ΛCDM), see Sec. III D.

Dataset	ΛCDM χ^2	ECF χ^2
Planck 2018 high-ℓ	2345.2	2347.0
Planck low-ℓ and lensing	420.1	420.1
BAO summary sector (BOSS DR12)[c]	6.2	6.2
Local H_0	29.41	0.10
Weak lensing S_8	12.07	0.00
Total effective χ^2	2816.18	2776.70

relative to the reference ΛCDM fit (historical value at $\Omega_m = 0.315$; NOT RECOMPUTED with the active $\Omega_m = \omega_m/h^2 \simeq 0.267$, which requires a dedicated ECF perturbation (Boltzmann) solver that does not yet exist). The dominant gains arise from the local H_0 and weak-lensing sectors, while the high-ℓ CMB contribution remains nearly neutral in this summary treatment. This near-neutrality is illustrated in Fig. 1; its robustness should ultimately be tested with a full ECF implementation in a Boltzmann solver.

The public tension-sector script associated with the companion analysis reproduces the six-point partial comparison built from local H_0, weak-lensing S_8, and BAO summary constraints. The Planck high-ℓ near-neutral contribution quoted in Table I belongs to the broader likelihood accounting described in the extended manuscript. The ECF sector introduces $k_{\rm ECF} = 2$ free parameters relative to the ΛCDM baseline ($F_{\rm ion}$, $\gamma_{\rm spin}$; historical count, retained unchanged although $\gamma_{\rm spin} = 0.3116$ is withdrawn as canonical calibration; H_0 is not counted separately, as it depends strictly on $F_{\rm ion}$ at fixed CMB acoustic angle, and $\tau_{\rm tor}$ is not varied in this six-dataset test), giving

$$\Delta\mathrm{AIC} \simeq -35.5, \qquad \Delta\mathrm{BIC} \simeq -23.8. \tag{11}$$

On the Kass–Raftery scale ($|\Delta\mathrm{BIC}| > 10$) these values would constitute very strong evidence in favour of the ECF over ΛCDM; they refer, however, to the historical configuration $\Omega_m = 0.315$ and have not been recomputed with the active $\Omega_m = \omega_m/h^2 \simeq 0.267$, which requires a dedicated ECF perturbation (Boltzmann) solver that does not yet exist. At the active value the BAO comparison is less favourable to the ECF (Sec. III D). These values are useful diagnostics of the adopted comparison, but they should not be read as a replacement for an eventual end-to-end likelihood analysis in a public Boltzmann pipeline. The BIC is quoted without a second decimal because its penalty term $k \ln N$ depends on a data-point count $N \approx 2500$ that is itself known only to within a few tens of points; a further digit would claim a precision the input does not carry.

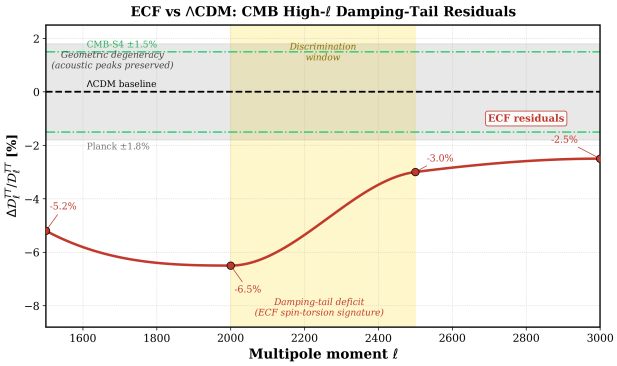


FIG. 1. **High-multipole CMB temperature residuals.** Comparison of illustrative C_ℓ^{TT} residuals for the calibrated ECF model and the ΛCDM baseline for $\ell > 1500$. The purpose of the figure is to show the approximate preservation of the acoustic peak structure in the benchmark implementation.

Reproduced by:
`Scripts_F1/plot_planck_tt_residuals_ecf.py`.

D. BAO Distances, Cosmic Age and the Open Trilemma

The reduced sound horizon does not by itself imply a direct conflict with BAO observations, because the standard BAO observables are ratios such as $D_M(z)/r_s$ and $H(z)r_s$, rather than an isolated measurement of r_s. In the calibrated ECF trajectory, the shift from the reference $r_s \simeq 147.1$ Mpc to $r_s^{\rm ECF} \simeq 135.65$ Mpc is accompanied by a corresponding increase of H_0 toward the local value, which partly compensates the change in these ratios. The compensation is not complete: with $\Omega_m = \omega_m/h^2 \simeq 0.267$, the four-bin diagonal comparison (BOSS DR12 + eBOSS DR16) gives $\chi^2_{\rm BAO} = 12.11$ for the ECF against 1.79 for ΛCDM ($\Delta\chi^2_{\rm BAO} \simeq +10.3$; e.g. $D_M(1.48)/r_s = 31.37$ against the eBOSS DR16 QSO BAO+RSD consensus 30.21 ± 0.79 [32, 33], a 1.5σ offset), so that the BAO sector currently disfavours the ECF background. The comparison uses $r_s^{\rm ECF}$ as the ruler of a measurement published in units of $r_{\rm drag}$ and is CALCULATED-CONDITIONAL on the open problem `PO-ECF-BAO-RDRAG`.

An important distinction applies to CMB peak observables: the ECF rescaling of r_s is compensated by a coherent increase of H_0, so the *angular positions* of the acoustic peaks (D_A/r_s) are preserved by construction *only at fixed* $\Omega_m = 0.315$; with $\Omega_m = \omega_m/h^2 \simeq 0.267$ and $w = -1$, θ_* falls by $\simeq 6.5\%$ and the peak positions are not preserved (PO-F1-THETA). The *heights* of the peaks, controlled by $\Omega_b h^2$, are not directly modified at the back-

ground level; their complete consistency check requires a full Boltzmann treatment (a dedicated Einstein–Cartan perturbation solver, deferred to Foundation III).

Cosmic chronometry, by contrast, is no longer a point of tension. With the strict $w = -1$ Foundation I ansatz and $\Omega_m = \omega_m/h^2 \simeq 0.267$, the benchmark predicts

$$t_0 \simeq 13.33 \text{ Gyr}. \tag{12}$$

This value lies above the globular-cluster lower bound $t_{\rm GC} = 13.32 \pm 0.10_{\rm stat} \pm 0.5_{\rm sys}$ Gyr and 0.17 Gyr below the age of the Universe inferred by the same authors [16], a chronometric compatibility check rather than a direct test of t_0 *(corrected 22/10/2026: formerly compared with the globular-cluster age as an estimate of t_0)*: the age tension reported in earlier versions (computed with $\Omega_m = 0.315$) disappears with this closure, and no dynamical $w(z)$ is required to meet the age bound. The value remains a falsifiable implication of simultaneously imposing the local H_0 scale, the contracted ECF sound horizon, and the adopted late-time equation of state. The difficulty is displaced to the BAO distances discussed above and to the (w_0, w_a) comparison with DESI (Sec. IV); it is not resolved here.

IV. OBSERVATIONAL TESTS

The value of the ECF construction depends on whether it leads to falsifiable consequences beyond parameter shifts. We therefore identify four observational directions where the framework can be tested.

One consequence should be stated before the others, since it is the most exposed. With $\Omega_m = \omega_m/h^2 \simeq 0.267$, the calibrated solution gives $t_0 \simeq 13.33$ Gyr for $w = -1$ (13.19 Gyr at the calibrated (w_0, w_a) point, i.e. 0.13 Gyr *below* the globular-cluster age, within its 0.5 Gyr systematic; 19/10/2026), above the globular-cluster lower bound $t_{\rm GC} = 13.32 \pm 0.10_{\rm stat} \pm 0.5_{\rm sys}$ Gyr and 0.17 Gyr below the age of the Universe $t_U = 13.5^{+0.16}_{-0.14}\,(\text{stat}) \pm 0.5\,(\text{sys})$ Gyr inferred by the same authors [16]; for $w = -1$ the age is therefore no longer the exposed point. The exposure has moved to the late-time distance sector: the BAO comparison of Sec. III D ($\Delta\chi^2_{\rm BAO} \simeq +10.3$) and the confrontation of the calibrated (w_0, w_a) trajectory with DESI DR1, whose χ^2 over the ten-observable subset of the companion calibration script — diagonal errors, not the global DESI DR1 χ^2 — rises to 55.3 (against 13.7 with $\Omega_m = 0.315$). Within the tanh-crossing family of effective $w(z)$ studied in the companion manuscript, no point satisfies the BAO and age constraints simultaneously (0 of 2500), so the H_0–Age–BAO trilemma remains irreducible in that family, independently of any age deficit; the broader problem remains open (PO-F2-5 of the companion manuscript [2]). The age-based falsification criterion stated in earlier versions no longer applies for $w = -1$. *The trilemma remains unresolved.*

A. Cosmic Birefringence

Macroscopic spin-torsion dynamics can generate parity-violating effective terms in the early universe. In the calibrated ECF implementation, this motivates a small isotropic rotation of the CMB polarization plane,

$$\beta_{\rm ECF} \simeq 0.35^\circ. \tag{13}$$

The predictive content is that the ECF yields a *non-zero* birefringence (unlike a pure cosmological constant); the specific magnitude 0.35° is a consistency check, the effective coupling being calibrated to the Minami–Komatsu anchor rather than derived ab initio (see the extended manuscript). This value is close to current birefringence inferences from Planck polarization analyses [17–19] ($\beta_{\rm Planck} = 0.30^\circ \pm 0.11^\circ$, 2.7σ, Planck PR4 [19] *(Corrected 22/10/2026: this value was previously attributed to Refs. [17, 18] alone)*; $\beta_{\rm ACT} = 0.215^\circ \pm 0.074^\circ$, ACT DR6 [20]). Future measurements by LiteBIRD and future CMB surveys *(DOE/NSF support for CMB-S4 ended in July 2025)* can therefore provide a sharp test of this sector [21, 22].

The same geometric framework admits an effective dark-energy equation of state, adopted as a calibration point: no deposited script produces it, and the $\Delta\chi^2$ budget contains no w_0 or w_a term; the deposited `calibrate_w0wa.py` gives $w_0 = -0.907 \pm 0.004$ (open problem PO-F1-W0 of the extended manuscript),

$$w_0 \simeq -0.904, \qquad w_a \simeq -0.153, \tag{14}$$

calibrated, not derived: w_a is set through $\xi = 1.645$, a parameter calibrated on the DESI preference used by the ECF so as to reproduce the target value of w_a, and is therefore not a prediction independent of DESI [2]. The present-day value satisfies the numerical relation $w_0 = -1 + \Omega^{\rm peak}_{\rm spin} = -0.907$ (matching -0.904 to 0.3%), which is an *empirical* identification rather than a derivation — whether the action implies such a link between the spin and torsion sectors is stated as an open problem in the companion Letter [1] — whereas the evolution parameter $w_a = -\xi\,\Omega^{\rm peak}_{\rm spin}$ retains a phenomenological factor $\xi \simeq 1.6$ that is *not* derived *ab initio* (script `calibrate_w0wa.py`; open problem PO-F1-B, see the companion extended manuscript). An independent MCMC comparison against the 12 DESI DR1 BAO observables, using the *full published covariance matrix* (Adame et al. 2024), gives a Mahalanobis distance of 2.03σ under the Planck prior ($h = 0.674$) and 6.98σ under the SH0ES prior ($h = 0.730$) [27, 28]. An earlier diagonal-errors check reported 0.78σ (SH0ES) with a caveat that full covariance "may shift these values by 10–30%"; that estimate is superseded — the shift is an order of magnitude larger. The two priors yield radically different CPL best-fits, reflecting the unresolved H_0 tension: the SH0ES prior is itself in strong tension with DESI BAO combined with BBN (the $\rm H_0DN$ value lies 5.0σ from DESI DR2 BAO+BBN [28, 29]), so the

SH0ES + DESI combination is not internally consistent, and the 6.98σ figure should be read in that light rather than as an isolated failure of the ECF prediction. In this consistency check the χ^2 of the ECF point is evaluated at fixed $(w_0, w_a) = (-0.904, -0.153)$ and at the Ω_m of the CPL best fit; it gives $\Delta\chi^2(\mathrm{ECF} - \Lambda\mathrm{CDM}) = +6.2$ under the Planck prior and $+38.6$ under the SH0ES prior (`plot_ecf_desi_mcmc_v3_0.py`) [2]. *(Corrected 16/10/2026: earlier versions quoted* -3.0 *at* $r_d^{\mathrm{ECF}} = 148$ *Mpc,* $\approx +20$ *and* $\Omega_m \approx 0.338$*, values produced by no deposited script.)* The MCMC posterior distance and the point $\Delta\chi^2$ measure different quantities: the former compares the ECF point to the marginalised posterior in the (w_0, w_a) plane (integrating over Ω_m), the latter a single point; the posterior distance is the more appropriate measure of (w_0, w_a)-plane compatibility [2]; the posterior itself is shown in Fig. 2.

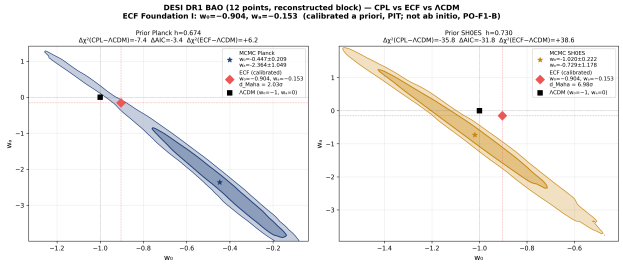


FIG. 2. **Calibrated ECF point vs DESI DR1 BAO posterior (two H_0 priors: Planck $h = 0.674$ and SH0ES $h = 0.730$).** Filled contours: 68% and 95% posterior in the (w_0, w_a) plane from an independent MCMC on 12 DESI DR1 BAO observables. Red diamond: ECF point $(w_0, w_a) = (-0.904, -0.153)$, *calibrated* (not an *ab initio* derivation; cf. PO-F1-B in the extended manuscript): w_a is set through $\xi = 1.645$, a parameter calibrated on the DESI preference used by the ECF, so the point is not a prediction independent of DESI; origin of the point: PO-F1-W0 [2]. Black square: ΛCDM. Using the full published covariance matrix (Adame et al. 2024, intra-bin D_M–D_H correlations), the Mahalanobis distance of the ECF point is 2.03σ under the Planck prior and 6.98σ under the SH0ES prior. The strong prior-sensitivity reflects a near-perfect degeneracy $\rho(w_0, w_a) \approx -0.99$, along which the ECF point sits perpendicular; the SH0ES prior is itself in strong tension with DESI BAO+BBN (5.0σ for the H_0DN value against DESI DR2 BAO+BBN [29]), so that combination is not internally consistent to begin with. Full analysis in Ref. [2].

Reproduced by: `Scripts_F1/plot_ecf_desi_mcmc_v3_0.py` (the F3 companion script draws the same contours; the F1 one carries the status labels of PO-F1-B).

B. CMB Damping-Tail Residuals

If the pre-recombination expansion history is modified, the photon diffusion scale and damping-tail structure can also be affected. The benchmark ECF solution forecasts a residual high-ℓ temperature-power deficit of order

$$\frac{\Delta D_\ell}{D_\ell} \sim 2.5\% - 6.5\%, \qquad 2000 \lesssim \ell \lesssim 3000. \tag{15}$$

This signature helps distinguish ECF-like scenarios from purely phenomenological shifts in r_s that leave diffusion physics unchanged.

C. Contracted Acoustic Ruler

The reduced sound horizon can also be probed by future 21 cm intensity-mapping observations. In the ECF benchmark,

$$r_s^{21\mathrm{cm}} \simeq 135.65 \pm 1.5 \text{ Mpc}, \tag{16}$$

so a direct or indirect reconstruction of the acoustic scale outside the standard supernova-distance-ladder context would provide an important consistency test.

D. Blue-Tilted Primordial Gravitational Waves

A torsion-driven bounce can motivate a blue-tilted stochastic gravitational-wave background, in contrast to the nearly scale-invariant tensor spectra associated with minimal slow-roll inflation.

The blue tilt of the predicted gravitational-wave background should not be read as a freely adjustable spectral index, but as the qualitative consequence of the post-bounce (expanding) stiff era, during which the spin contribution scales as $\rho_{\mathrm{spin}} \propto a^{-6}$ and temporarily dominates the high-density regime. In this background, high-frequency tensor modes are amplified more efficiently than long-wavelength ones, producing a spectral energy density that rises toward high frequency before terminating at a cutoff scale set by the minimum scale factor $a_\star$. The representative values quoted below — $\Omega_{\mathrm{GW}}h^2 \sim 10^{-12}$ near millihertz frequencies and $a_{\min} \sim 10^{-32}$ (the cutoff itself lies near 10^{11}–10^{12} Hz; the millihertz amplitude is not computed) — should therefore be understood as a phenomenological target range for future transfer-function calculations, not as a precision forecast: the complete derivation of the tensor spectral index n_T, of the mode-matching across the bounce, and of the detector-level signal shape is deferred to the perturbative treatment developed in Foundations II and III [3, 4].

a. Scalar spectrum: an inherited initial condition. The blue tensor tilt above concerns the *tensor* sector only. The *scalar* spectrum is not predicted here: the cosmological modes relevant to CMB observations were super-horizon throughout the stiff era and their amplitudes were dynamically frozen well before recombination. Because the spin contribution scales as a^{-6} and becomes negligible long before those modes re-enter the Hubble radius, they are effectively insensitive to the local spin-fluid physics of the bounce. The standard nearly

scale-invariant initial conditions ($n_s \simeq 0.965$) required by CMB observations are therefore **inherited** by the ECF late-time universe rather than derived from it, and n_s enters the present analysis as an *input*, on the same footing as Ω_b or A_s (see the extended version [2] and Foundation III — Cosmological Physics).

b. Terminological note — two distinct stiff eras. Throughout this paper, "stiff era" denotes the *transient, post-bounce, pre-recombination* phase during which the Weyssenhoff spin fluid temporarily dominates ($\Omega_{\rm spin}^{\rm peak} \approx 9.3\%$) and reduces the sound horizon — the mechanism underlying the H_0 result of this work, denoted $N_{\rm stiff}^{\rm exp}$. It should not be confused with the *contracting* stiff phase that precedes the bounce, $N_{\rm stiff}^{\rm contr}$, discussed in Foundation III, which governs the horizon problem and the primordial spectrum. The two eras lie on opposite sides of the bounce and play different roles; where an e-fold count is quoted, the relevant era must be specified.

The extended manuscript uses representative values of order $\Omega_{\rm GW}h^2 \sim 10^{-12}$ near millihertz frequencies and $a_{\rm min} \sim 10^{-32}$ to illustrate the relevant scale. These numbers should be interpreted as a target range for future transfer-function calculations, not as a precision prediction of the short paper.

V. HIGH-REDSHIFT STRUCTURE

The transient stiff-like phase may also influence the initial conditions for early structure formation. Earlier versions attributed an effective early growth enhancement to the pre-recombination dynamics; that linear boost is not derived and is withdrawn as a mechanism (open problem PO-F1-BOOST of the extended manuscript). The mechanism retained is seeding by Macro-Knots, whose abundance and collapse efficiency are not derived. At the Eddington rate from $z = 20$, a $10^5\,M_\odot$ seed reaches the black holes UHZ1 ($z = 10.1$) [23, 24] and J1342+0928 ($z = 7.54$) [25] at $f_{\rm Edd} \simeq 1.1$–1.3, where a $10^2\,M_\odot$ stellar seed needs $\simeq 1.9$–2.8; the absence of compact objects of $\gtrsim 10^5\,M_\odot$ at $z \gtrsim 10$ would falsify the mechanism. The extended manuscript further discusses primordial Macro-Knot seeds, with characteristic masses of order 10^4–$10^6 M_\odot$, as possible topological initial conditions formed around the Great Annihilation freeze-out (e^+e^- annihilation epoch, $T \sim 1$ MeV) — not the much hotter QCD transition, which sets the smaller Micro-Knot population instead. Earlier versions also compared the ECF with an abundance target $n(M > 10^{10}M_\odot, z = 15) \simeq (2$–$4) \times 10^{-4}\,{\rm Mpc}^{-3}$ through a modified Press–Schechter chain; the target has no traceable JWST source (it is not in Labbé *et al.* [26]; PO-F1-JWST-SRC of the extended manuscript) and the chain used a variance too low by a factor $\simeq 2$ (PO-F1-SIGMA-M), so both are withdrawn *(17/10/2026). (Corrected 16/10/2026: earlier versions gave the redshift as $z = 10$.)* The direction of the effect is that the ECF background could ease the formation of massive high-redshift systems. However, JWST stellar-mass estimates remain sensitive to photometric modeling, star-formation histories, dust, and selection effects [26]. We therefore present this point as consistency with a possible high-redshift trend, not as an independent confirmation of the model. The full dark-sector and Macro-Knot construction belongs to the companion Foundation II analysis and is only summarized here to maintain continuity of the program.

VI. REPRODUCIBILITY

The numerical code, dataset preprocessing scripts, and master orchestration script used in the companion analysis are made publicly available:

https://github.com/pfichant/spin-torsion-cosmology

The repository includes environment configuration files intended to support reproducibility. The extended manuscript documents which scripts correspond to each numerical result and figure [2].

VII. CONCLUSION AND OUTLOOK

We have presented a compact Einstein–Cartan cosmological framework in which the spin density of matter induces a repulsive contribution to the early Friedmann dynamics, regularizing the high-density regime through a non-singular bounce. The transient stiff-like phase before recombination is carried by a distinct calibrated positive a^{-6} branch, not derived continuously from that repulsive correction (PO-F1-SIGMA0). Its central calibration is the effective amplitude $F_{\rm ion} = 1.2765$, which sets the sound-horizon reduction and $H_0 \simeq 73.04\ {\rm km\,s^{-1}\,Mpc^{-1}}$, the latter conditional on θ_* being preserved, which holds only at fixed $\Omega_m = 0.315$ (PO-F1-THETA). With the closure $\Omega_m = \omega_m/h^2$, the same background lowers S_8 toward the weak-lensing range (CAMB ΛCDM reference $S_8 \simeq 0.780$); the residual correction to $S_8 = 0.766$ and the growth role of the spin sector, of which $G_{\rm eff}(k)$ is a phenomenological parametrisation, await a dedicated Einstein–Cartan perturbation solver. In the historical configuration, the calibrated solution improved the effective global χ^2 by $\Delta\chi^2 \simeq -39.5$ (value computed with $\Omega_m = 0.315$; not recomputed with $\Omega_m = \omega_m/h^2$ — requires a dedicated ECF perturbation (Boltzmann) solver, which does not yet exist) for the dataset combination of the companion analysis. Several aspects remain provisional: a full Boltzmann-code implementation, a perturbative ECF treatment of scalar and tensor modes, and a systematic likelihood analysis are required before the framework can be assessed at the level of established models. Its defining feature is falsifiability, through the observational channels identified above; precise null results would strongly constrain the geometric interpretation.

The present paper is restricted to the metric and background phenomenology of Foundation I within the broader program outlined in the PIT [1]. The chiral and topological consequences of the torsion-driven bounce — baryogenesis, geometric dark-sector structures, and residual cosmic alignment — are developed in the companion Foundation II manuscript [3], and the strong-gravity sector (singularity avoidance in black hole interiors, transient Einstein–Rosen geometry, compact-object observables) in Foundation III [4].

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